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Module : Research Question Recent

Q1. What is WSL for ? How is it related Linux as an Operating System ?

An Operating System is the main software that controls your computer

For example :

Windows	Operating System
Linux	Operating System
Android	Operating System
Mac OS	Operating System

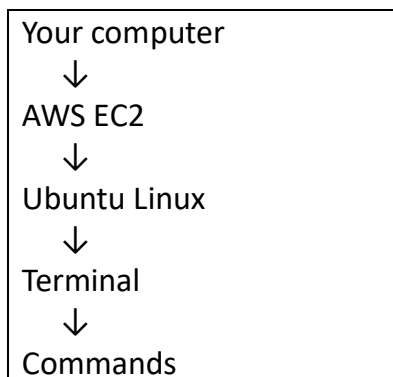
Application such as Chrome , VS Code and Docker and etc run on top and withing the operating system

Linux is an operating system based on the Linux kernel and some of the Linux distributions are :

Ubuntu
Amazon Linux

If you use an AWS EC2 instance running ubuntu , you are using a Linux operating system

For example :



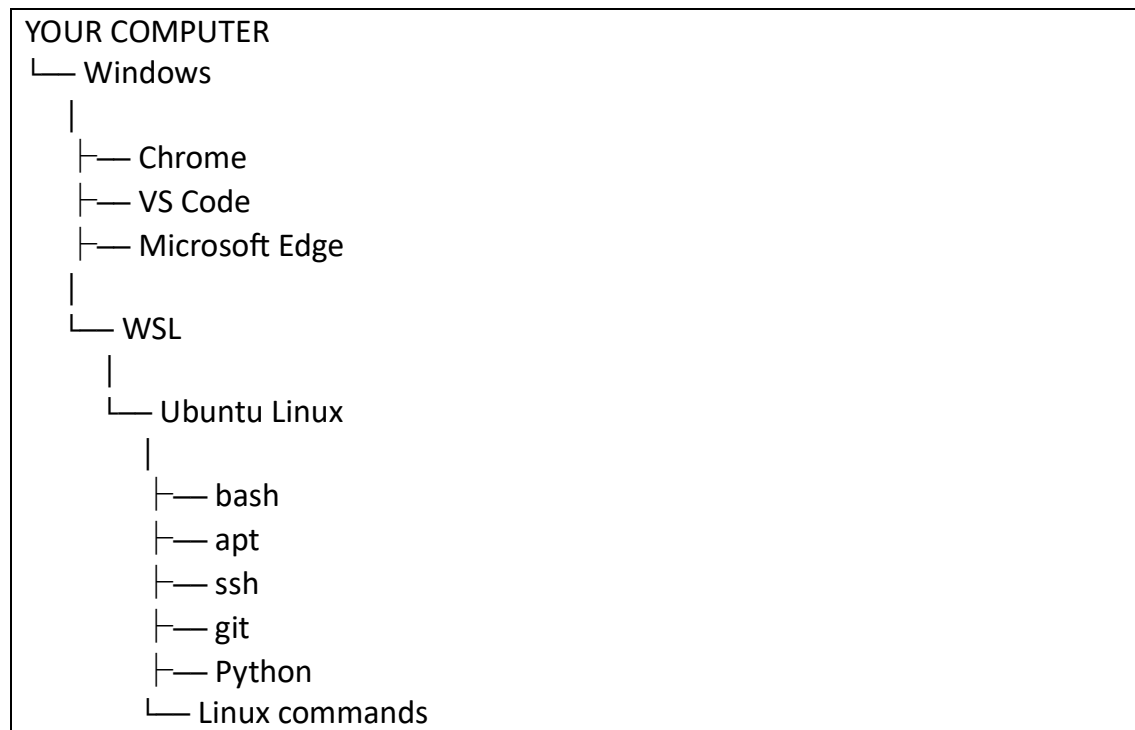
When you type the following commands : You are interacting with Linux

ls
cd
sudo apt update

WLS is known as Windows Subsystem for Linux and it allows you to run a Linux environment inside Windows

So instead of installing Ubuntu as your entire computer operating system , you can use Ubuntu from inside Windows

For example :



Suppose we have Windows and want to work with Linux tools and instead of removing windows and putting Linux OS whole

WSL gives you a Linux Workspace inside your Windows and you can switch between them whenever you want

Without WSL

Windows PC



Windows only

With WSL

Windows PC



Windows



Linux environment

IS WSL itself Linux :

WSL is not a Linux distribution and it is a Windows feature or subsystem that provides a way to run Linux environments

For example :

Windows



└ WSL



└ Ubuntu



└ Linux environment

So Windows is your main OS and WSL is the technology provided by Windows for Linux environments and Ubuntu is the Linux distribution you can install inside WSL

Happens when you open ubuntu in WSL :

Suppose you install Ubuntu through WSL and open Ubuntu and see the

username@computer:~\$

And you can type : and see :

ls
Documents
Downloads
Projects
Test

Windows user need Linux :

A huge amount of server infrastructure uses Linux and for example :

AWS EC2
↓
Ubuntu Linux
↓
Nginx
↓
Docker
↓
Jenkins
↓
Spring Boot

But your personal computer might be Windows 11 but AWS server could be Ubuntu Linux

So need use Linux commands and WSL gives you Linux environment on your Windows computer to practice Linux

So therefore , WSL known as Windows Subsystem for Linux is a Windows feature that lets you run a Linux environment such as Ubuntu directly alongside Windows and mainly so you can use Linux tools , commands and development workflows without replacing Windows

Q2. What is the difference between Linux and Unix ?

In a computer we have the following :

You	User
Applications	Docker , Jenkins , Java , Browser
OS	Linux , Unix , Windows
Hardware	CPU , RAM , disk , network card

Unix is a family of operating system that started in the 1970S AT&T Bell Labs and used for servers and systems for examples :

IBM
Oracle Solaris
HP
MacOS

Think of Unix as an older and original family of operating systems

Linux is an operating system kernel and its like Unix meaning it follows many ideas and behaviour's associated with Unix but Linux itself is not Unix

Examples of Linux Distributions :

Ubuntu
Red Hat Enterprise Linux
Amazon Linux
Kali Linux

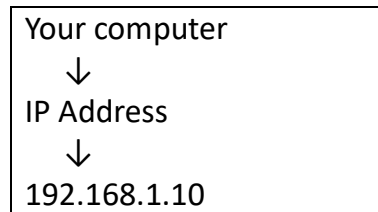
So Unix is an original design or concept and Linux is a different system built following many of those Unix ideas

Linux behaves very similar to Unix , but Linux was developed separately

Q3. Public IP and Private IP

An IP address is like an address for a device on a network and computer has an IP address so network traffic knows where to go

For example :



A public IP is an IP address that can be used to communicate over the internet and can access it from your laptop for example :

13.234.56.78

Imagine you AWS server has :

Public IP = 13.234.56.78

Can access this from your laptop and internet can reach that public IP



A Private IP is used inside a private network and these address are not directly reachable from the public internet and example :

10.0.1.10
10.0.1.20
192.168.1.10

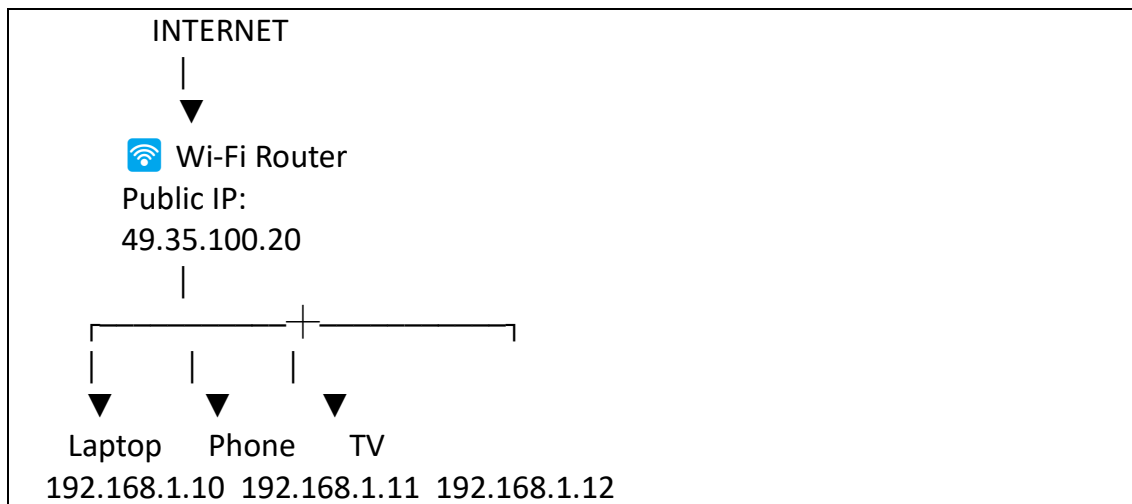
10.0.0.0 - 10.255.255.255

172.16.0.0 - 172.31.255.255

192.168.0.0 - 192.168.255.255

Home Wi-Fi Example

Suppose your home has :



Your laptop might have different Private IP such as : Laptop , Phone and TV while connecting to router which has Public IP

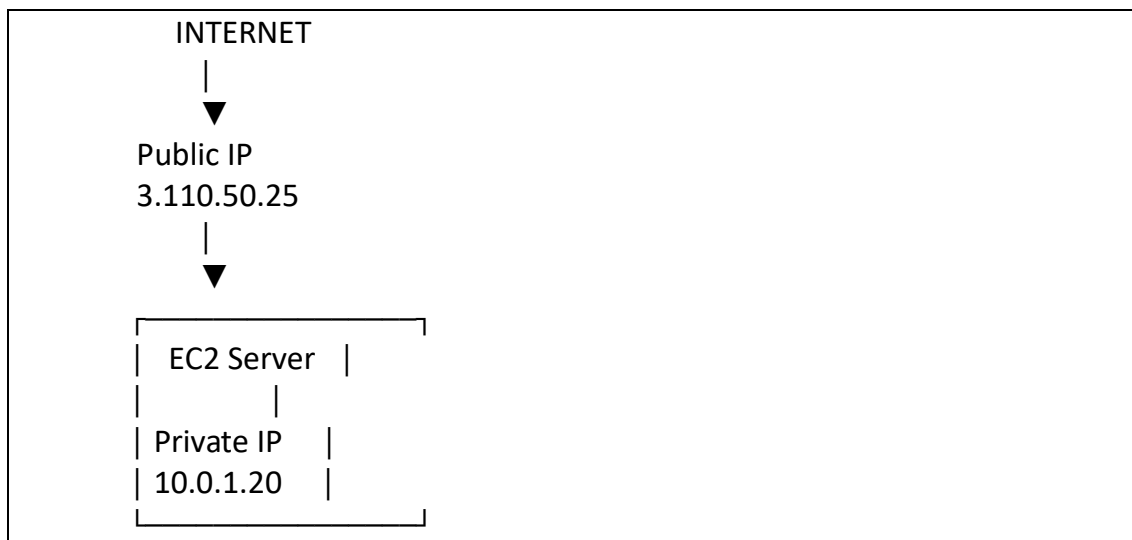
When your laptop accesses Google :



Your laptop private IP doesn't directly travel across the public internet as the router performs NAT which is Network Address Translation

So public IP used for communication involving the internet and private IP is used for communication inside the AWS private network

Imagine you create an EC2 inside a VPC such as



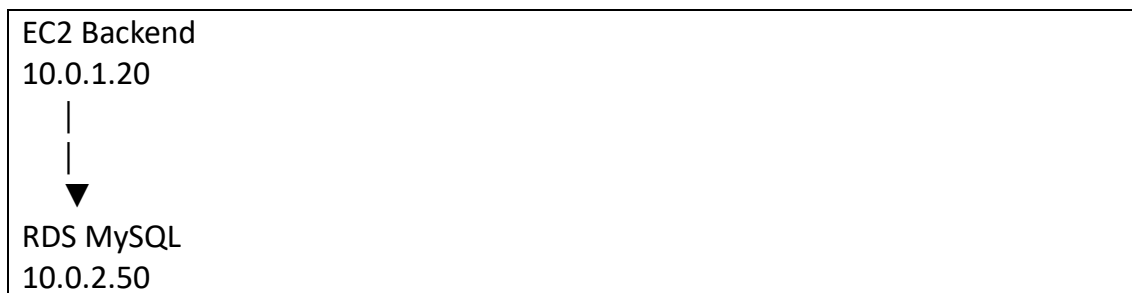
The EC2 instance can have

Public IP = 3.110.50.25
Private IP = 10.0.1.20

So in Public IP :



So in Private IP :



So in backend can communicate with the database using the database private networking using Private IP

Public IP vs Private IP :

Feature	Public IP	Private IP
Used on internet	Yes	No
Used inside private network	Can be	Yes
Globally Unique	Yes	No
Example	54.200.10.50	10.0.1.20
Internet can reach	Yes	No
AWS	Interfacing EC2 / ALB	EC2 - RDS
Security	Exposed	Isolated

Q4 . Application Server, Webserver - what are these - Name some examples for Java and Python.

A server is a computer or program that waits for requests and gives responses and also first request is made and the request is processed and final gives the result

So user gives request the server and the server processes it and gives the response to the user

A web server mainly handles HTTP / HTTPS requests and serves web content and common jobs are :

Receive HTTP requests
Serve HTML and CCS , Java Script and images and files
Reverse proxy requests
Handles SSL / HTTPS

So browser uses nginx to create index.html and connect it with browser and nginx returns index.html , style.css . app.js and log.png which we created and this displays in the browser

An application server runs your application actual business logic as web server receives and deliver web traffic while application server runs the application

Suppose for Electricity Bill Application , the user requests get the data of bills with no 101

GET /api/bills/101

So the application does this known as application logic

Requests the request
Check customer ID
Query MYSQL
Calculates
Apply business rules
Return JSON

In Java the spring boot is used for Java backend applications such as Tomcat as it's an embedded web server

So some know tomcat as a web server or an application server but actually tomcat is a java server or runtime that can receive HTTP requests and run Java Web Applications

In Spring boot application , Tomcat is often embedded inside your application such as :

```
java -jar electricity-bill.jar
```

When you run this , Spring boot application starts tomcat internally with :

```
Tomcat → Port 8080
```

And can access using

```
https://localhost:8080
```

And also Nginx receives the request first such as :

```
https://mywebsite.com
```

So Nginx can decide :

```
/api/* → Spring Boot :8080
```

```
/static/* → Frontend files
```

Python is known for web application frameworks and frameworks such as Django , Flask and Fast API

JAVA vs Python

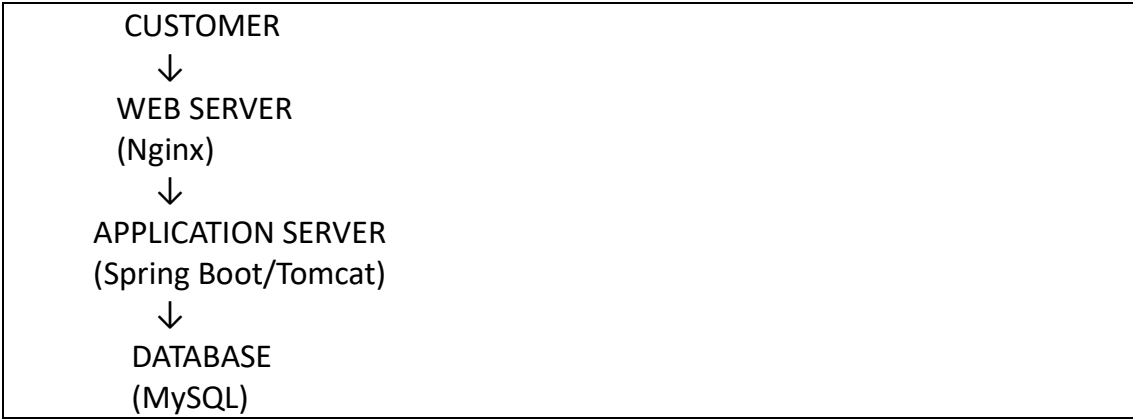
Purpose	Java	Python
Framework	Spring boot	Django
Framework	Spring boot	Flask
Framework	Sprin boot	Fast API
Server / Runtime	Tomcat	Gunicorn
Web Server	Nginx	Nginx
Web Server	Apache	Apache

So , Spring boot , Django , Flask , FAST API are application framework

Tomcat are application servers / server runtimes

Nginx , Apache HTTP Server are web servers

So :



Static vs Dynamic :

Static are things that don't require application processing and Nginx can server these directly using browser to Nginx and final index.html

index.html
style.css
app.js
logo.png

Browser
↓
Nginx
↓
index.html

Dynamic :

Things that require application logic and these go to your application such as Browser to Nginx to Spring boot and finally MSQL

GET /api/bills/101
POST /api/login
POST /api/payment
GET /api/customers

Browser
↓
Nginx
↓
Spring Boot
↓
MySQL

So therefore ,

- **Nginx** → Web server / reverse proxy
- **Spring Boot** → Java application
- **Tomcat** → commonly embedded inside Spring Boot
- **MySQL RDS** → Database
- **Docker** → Packages/runs your application
- **ALB** → AWS load balancer

Web Server who receives the web traffic

Application server / runtime runs my application / business logic

Database is who stores my data

Q5. what is Open source system ?

1. What is "free" in software system mean?
2. Is Open SORce system always "free" ? does "free" mean always money/dollar?

Concept	Meaning	Example
Open source	Source code is available and the license gives rights to use , study , modify and share	Linux , Jenkins , Nginx
Free of cost	No need to pay money	Free software
Free as freedom	Freedom / Permissions to use , modify or share according to the license	Modify open source code
License	Rules how software can be used , modified and shared	MIT , Apache
Can Open source cost money	Yes	Paid support , hosting , enterprise services
Is free always open source	No	A free app can have closed source
Does open source mean 0 dollars	No	Open source software uses paid services

AWS example	Open source software can be free while AWS infrastructure costs money	Jenkins on EC2
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